# DIGITAL LOGIC DESIGN

Lecture 05

Simplify the following Boolean expression:

$$\overline{A}BC + A\overline{B}\,\overline{C} + \overline{A}\,\overline{B}\,\overline{C} + A\overline{B}C + ABC$$

## Solution

**Step 1:** Factor $BC$ out of the first and last terms.

$$BC(\overline{A} + A) + A\overline{B}\,\overline{C} + \overline{A}\,\overline{B}\,\overline{C} + A\overline{B}C$$

**Step 2:** Apply rule 6 ($\overline{A} + A = 1$) to the term in parentheses, and factor $A\overline{B}$ from the second and last terms.

$$BC \cdot 1 + A\overline{B}(\overline{C} + C) + \overline{A}\,\overline{B}\,\overline{C}$$

**Step 3:** Apply rule 4 (drop the 1) to the first term and rule 6 ($\overline{C} + C = 1$) to the term in parentheses.

$$BC + A\overline{B} \cdot 1 + \overline{A}\,\overline{B}\,\overline{C}$$

**Step 4:** Apply rule 4 (drop the 1) to the second term.

$$BC + A\overline{B} + \overline{A}\,\overline{B}\,\overline{C}$$

**Step 5:** Factor $\overline{B}$ from the second and third terms.

$$BC + \overline{B}(A + \overline{A}\,\overline{C})$$

**Step 6:** Apply rule 11 ($A + \overline{A}\,\overline{C} = A + \overline{C}$) to the term in parentheses.

$$BC + \overline{B}(A + \overline{C})$$

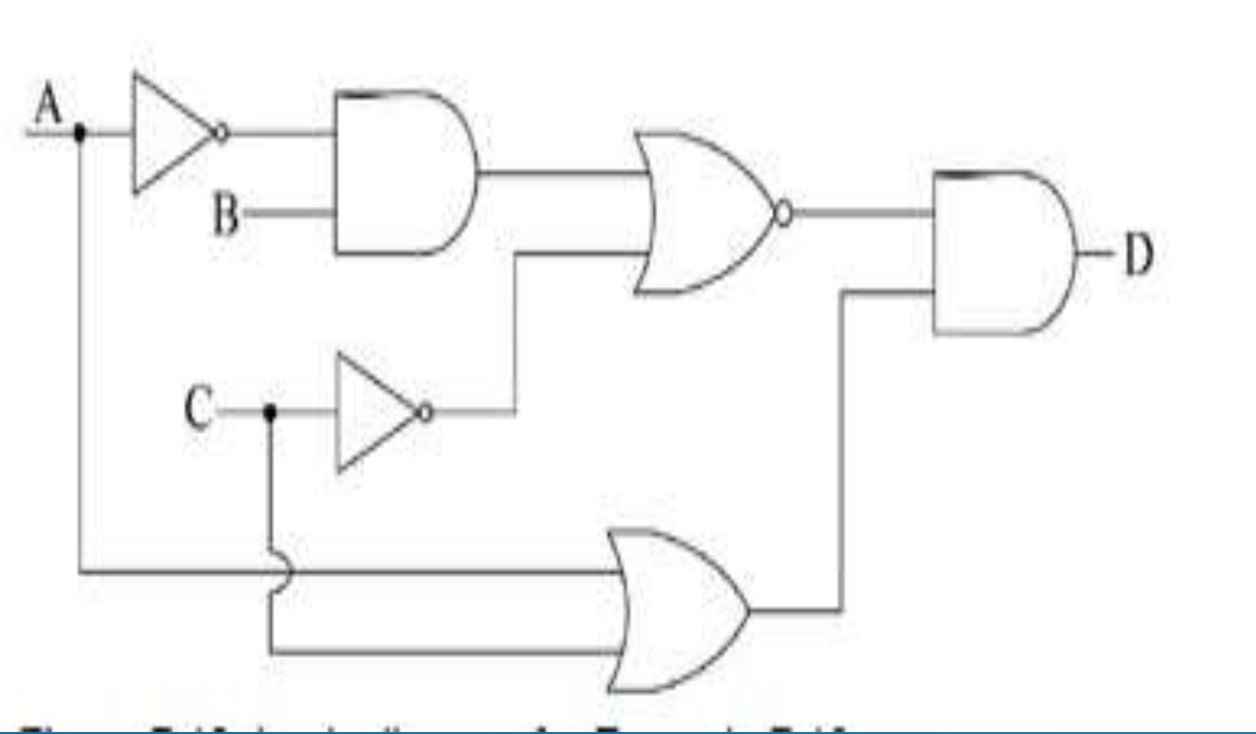
**Step 7:** Use the distributive and commutative laws to get the following expression:

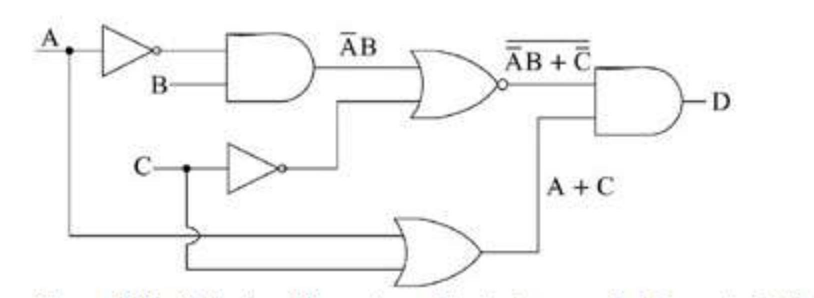
$$BC + A\overline{B} + \overline{B}\,\overline{C}$$

Simplify the Boolean expression and draw logic diagram.

$$
\begin{aligned}
\overline{(AC + \overline{B\overline{C}} + D)} + \overline{\overline{BC}} &= (\overline{AC})(\overline{\overline{B\overline{C}}})\overline{D} + \overline{\overline{BC}} && \text{Apply DeMorgan} \\
&= (\overline{AC})(B\overline{C})\overline{D} + BC && \text{Cancel double complements} \\
&= (\overline{A} + \overline{C})B\overline{C}\,\overline{D} + BC && \text{Apply DeMorgan and factor} \\
&= \overline{A}B\overline{C}\,\overline{D} + B\overline{C}\,\overline{D} + BC && \text{Distributive law} \\
&= B\overline{C}\,\overline{D}(1 + \overline{A}) + BC && \text{Factor} \\
&= B\overline{C}\,\overline{D} + BC && \text{Rule: } 1 + A = 1
\end{aligned}
$$

For the logic circuit of Figure. Find D = f(A, B, C), that is, express the output D in terms of the inputs A, B , and C. If possible, simplify the Boolean expression obtained, and implement it with a simplified logic diagram.



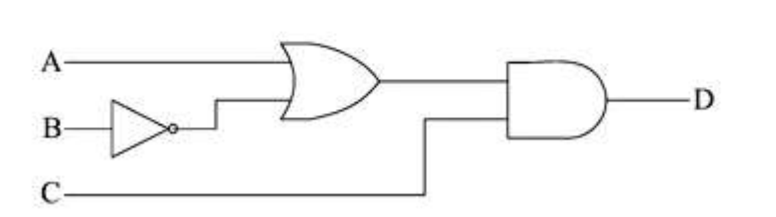



$$D = \left(\overline{\bar{A}B + \bar{C}}\right)(A + C) = \left(\overline{\bar{A}B}C\right)(A + C) = \left[\left(A + \bar{B}\right)C\right](A + C)$$

$$= \left(A + \bar{B}\right)(AC + CC) = \left(A + \bar{B}\right)(AC + C) = AAC + AC + A\bar{B}C + \bar{B}C$$

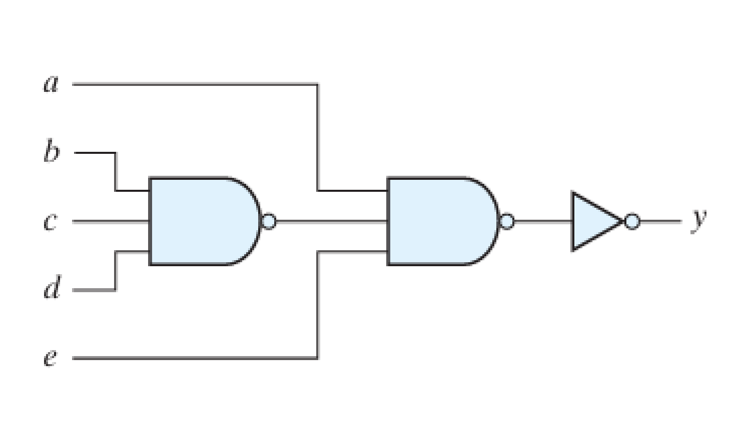
$$= AC + AC + \bar{B}C(A + 1) = AC + \bar{B}C(1) = AC + \bar{B}C$$

$$= \left(A + \bar{B}\right)C$$




| A | B | C | D |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 |

- Write Boolean expressions describing the outputs of the circuits described by the logic diagrams in Fig.



To find the Boolean expression and truth table for the given logic circuit, we will analyze the operation of each gate from left to right.

### 1. Boolean Expression Analysis

The circuit consists of three main stages:

1. **Stage 1 (NAND Gate):** A 3-input NAND gate takes inputs $b$, $c$, and $d$. Its output is:
   $\text{Output}_1 = \overline{b \cdot c \cdot d}$
2. **Stage 2 (NAND Gate):** Another 3-input NAND gate receives input $a$, the output from the first gate ($\text{Output}_1$), and input $e$. Its output is:
   $\text{Output}_2 = \overline{a \cdot (\overline{b \cdot c \cdot d}) \cdot e}$

3. **Stage 3 (NOT Gate):** An inverter (NOT gate) complements the result of the second NAND gate to produce the final output $y$:
   $y = \overline{\mathrm{Output}_2} = \overline{\overline{a \cdot (\overline{b \cdot c \cdot d}) \cdot e}}$

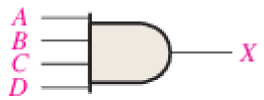
Using the **Double Negation Law** ($\overline{\overline{A}} = A$), the expression simplifies to:

$$y = a \cdot (\overline{b \cdot c \cdot d}) \cdot e$$

Alternatively, by applying **De Morgan's Law** to the middle term:

$$y = a(\bar{b} + \bar{c} + \bar{d})e = ae\bar{b} + ae\bar{c} + ae\bar{d}$$

Write Boolean expressions and construct the truth tables describing the outputs of the circuits described by the logic diagrams in Fig.



(a)

# STANDARD FORMS OF BOOLEAN EXPRESSIONS

All Boolean expressions, regardless of their form, can be converted into either of two standard forms: the sum-of-products form or the product-of sums form. Standardization makes the evaluation, simplification, and implementation of Boolean expressions much more systematic and easier.

# THE SUM-OF-PRODUCTS (SOP)

- The Sum-of-Products (SOP) Form When two or more product terms are summed by Boolean addition, the resulting expression is a sum-of-products (SOP).
- Some examples are:
  - AB + ABC
  - ABC + CDE + BCD
  - AB + BCD + AC
- Also, an SOP expression can contain a single-variable term, as in A + ABC + BCD.

# THE STANDARD SOP FORM

- So far, you have seen SOP expressions in which some of the product terms do not contain all of the variables in the domain of the expression.
- For example, the expression ABC + ABD + ABCD has a domain made up of the variables A, B, C. and D. However, notice that the complete set of variables in the domain is not represented in the first two terms of the expression; that is, D or D is missing from the first term and C or C is missing from the second term.

- **A standard SOP expression is one in which all the variables in the domain appear in each product term in the expression. For example, ABCD + ABCD + ABCD is a standard SOP expression.**

# THE PRODUCT-OF-SUMS (POS)

- A sum term was defined before as a term consisting of the sum (Boolean addition) of literals (variables or their complements). When two or more sum terms are multiplied, the resulting expression is a product-of-sums (POS).
- Some examples are $(A + B)(A + B + C)$ $(A + B + C)(C + D + E)(B + C + D)$ $(A + B)(A + B + C)(A + C)$

# THE STANDARD POS FORM

- So far, you have seen POS expressions in which some of the sum terms do not contain all of the variables in the domain of the expression.

- For example, the expression $(A + B + C)\ (A + B + D)\ (A + B + C + D)$ has a domain made up of the variables A, B, C, and D.

- A standard POS expression is one in which all the variables in the domain appear in each sum term in the expression. For example, $(A + B + C + D)(A + B + C + D)(A + B + C + D)$

# CONVERTING PRODUCT TERMS TO STANDARD SOP

- Each product term in an SOP expression that does not contain all the variables in the domain can be expanded to standard SOP to include all variables in the domain and their complements. As stated in the following steps, a nonstandard SOP expression is converted into standard form using Boolean algebra rule 6 (A + A = 1) : A variable added to its complement equals
- 1. Step 1. Multiply each nonstandard product term by a term made up of the sum of a missing variable and its complement. This results in two product terms. As you know, you can multiply anything by 1 without changing its value.
- Step 2. Repeat Step 1 until all resulting product terms contain all variables in the domain in either complemented or complemented form. In converting a product term to standard form, the number of product terms is doubled for each missing variable.

# EXAMPLE

- Convert the following Boolean expression into standard SOP form: ABC + AB + ABCD
- **Solution** The domain of this SOP expression A, B, C, D. Take one term at a time. The first term, ABC, is missing variable D or D, so multiply the first term by (D + D) as follows: ABC = ABC(D + D) = ABCD + ABCD
- In this case, two standard product terms are the result. The second term, AB, is missing variables C or C and D or D, so first multiply the second term by C + C as follows: AB = AB(C + C) = ABC + ABC